

Confidently

Attack / Threat	Explanation from the chapter	Simple Example
Stealing Password Files	Obtaining passwords or login credentials without authorization. The slides mention this occurring through public Wi-Fi or by injecting a keylogger into a browser.	An attacker uses a keylogger to capture your username and password when you log into your university account.
Port Scanning	Sending a series of messages to a computer to learn which network services are available, each associated with a known port number.	An attacker scans a server and discovers that port 22 (SSH) and port 80 (HTTP) are open.
Shoulder Surfing	Spying on someone using an ATM or another electronic device to obtain information such as their PIN or password .	Someone watches over your shoulder while you enter your ATM PIN.
Eavesdropping	An electronic attack where digital communications are intercepted by someone they were not intended for . The slides associate this with Man-in-the-Middle (MITM) .	An attacker positions themselves between you and a service and intercepts information being transmitted.
Sniffing	Intercepting data by capturing network traffic using a <i>sniffer</i> , an application designed to capture network packets.	An attacker captures packets traveling across a network and examines them for sensitive information.
Privilege Escalation	A network intrusion that exploits programming errors or design flaws to give an attacker elevated access to a network, its data, and applications.	A normal user exploits a vulnerability and gains administrator privileges, allowing access to confidential files.

Human Errors / Mistakes	Unintentional human actions that can expose confidential information.	An employee accidentally emails a confidential document to the wrong person.
Intentional Damage / Harm	Intentional actions by a person that compromise confidentiality.	An employee deliberately gives confidential company information to an unauthorized person.

Method for Ensuring Confidentiality	Explanation	Simple Example
Data Encryption	Converts data into a protected form so unauthorized people cannot understand it.	Encrypting sensitive files stored on a laptop.
User IDs & Passwords	Uses login credentials to restrict access to authorized users.	A student must enter their university username and password to access their account.
Two-Factor Authentication (2FA)	Requires an additional form of verification beyond the normal login credentials.	Entering a password and then a verification code.
Biometric Verification	Identifies a person using distinguishing biological traits such as fingerprints, retina/iris patterns, voice, DNA, or signatures.	Using a fingerprint to unlock a device.

Security Tokens	A small hardware device carried by the owner to authorize access to a network service.	Using a physical authentication token when logging into a secure system.
Soft Tokens	A software-based security token that generates a single-use login PIN .	An authentication app generates a one-time PIN for login.
Key Fobs	A small programmable hardware device that provides access to a physical object requiring two-factor or multifactor authentication .	Using a key fob as an additional authentication factor to gain access.
Air-Gapped Computers	Isolates a computer or network so it cannot establish an external connection. Used as an extra measure for extremely sensitive information.	Keeping classified documents on a computer with no Internet/network connection.
Disconnected Storage Devices	Stores extremely sensitive information on storage devices that remain disconnected from networks.	Keeping confidential files on an external drive that is disconnected when not being used.

Integrity

Attack / Threat	Explanation	Simple Example
Viruses	Malicious programs can alter, corrupt, or delete data, causing it to lose its accuracy or reliability.	A virus modifies files stored on a computer, changing their contents.

Logic Bombs	Malicious code designed to execute when a particular condition or event occurs, potentially modifying or destroying data.	Malicious code is programmed to delete company files on a specific date.
Unauthorized Access	Someone without permission gains access and changes information or systems.	A student gains unauthorized access to the grading system and changes a grade.
Errors in Coding	Programming mistakes can cause data to be processed, stored, or modified incorrectly.	A bug in a banking application incorrectly changes a customer's account balance.
System Backdoors	A hidden or alternative method of accessing a system can bypass normal security controls, allowing unauthorized changes.	An attacker uses a backdoor to enter a system and modify stored records.

Method for Ensuring Integrity	Explanation	Simple Example
File Permissions & User Access Controls	Controls who is allowed to access or modify data.	Only authorized instructors can modify student grades.
Version Control	Helps prevent erroneous changes or accidental deletion by authorized users from becoming a problem.	Using Git to restore an earlier version of a file after an incorrect change.

Checksums / Hashing	Generates a value that can later be used to verify that data has not been changed .	Comparing a downloaded file's hash with the original hash to detect modification.
Backups / Redundancies	Allows affected data to be restored to its correct state.	Restoring an unmodified backup after data has been corrupted.
Cryptographic Checksums	Assigns a mathematical value (checksum/hash) to a file so it can later be tested for malicious changes.	A file's current hash differs from its original hash, indicating the file was modified.

Availability

Threat to Availability	Explanation	Simple Example
Device Failure	Hardware failure prevents authorized users from accessing information or systems.	A server's hard drive fails, making the system unavailable.
Software Error	A software problem or failure causes a system or service to become unavailable.	A bug causes a university website to crash.
Environmental Issues	Physical/environmental events such as heat, flooding, or power loss can make systems unavailable.	A power outage shuts down a company's servers.

Denial-of-Service (DoS) Attack	An attack that disrupts a service so legitimate users cannot access it.	An attacker overwhelms a website with requests until legitimate users cannot access it.
Network Intrusions	Unauthorized intrusion into a network can disrupt systems and prevent legitimate access.	An attacker compromises a network and causes its services to become unavailable.

Method for Ensuring Availability	Explanation	Simple Example
Backups	A backup copy can be stored in a geographically isolated location to prevent data loss.	A company keeps a backup of its data at a separate location in case the main server is damaged.
Fireproof / Waterproof Safe	Backup data can be physically protected against environmental threats.	Backup drives are stored in a fireproof and waterproof safe.
Firewalls	Extra security equipment or software can help protect systems against downtime and unreachable data.	A firewall blocks harmful network traffic that could disrupt a server.
Proxy Servers	Can help protect systems and services against disruptions that could make data unreachable.	A proxy server helps control network traffic reaching an organization's systems.

Web Application Firewall (WAF)	Protects web applications from attacks that could affect their availability. The slide gives Cloudflare as an example.	Cloudflare protects a website against malicious traffic that could make it unavailable.
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